

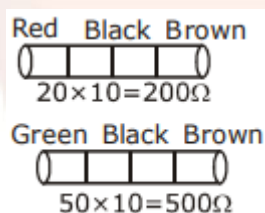
JEE Mains 8th April, 2019 Morning

Physics

1. A 200Ω resistor has a certain color code. If one replaces the red color by green in the code, the new resistance will be:
 A. 500Ω B. 400Ω
 C. 100Ω D. 300Ω

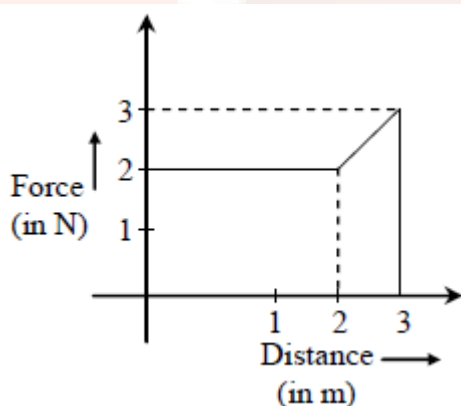
Ans. A

Explanation:



Red $\rightarrow$ Green $\Rightarrow 200\Omega \rightarrow 500\Omega$

2. A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3m is :



- A. 2.5 J B. 6.5 J
 C. 4 J D. 5 J

Ans. B

Explanation:

Area under force-displacement graph gives the value of work done.

Work = = Area under F–x graph= $\Delta K.E$

$$\Delta K.E = 4 + 2 + 0.5 = 6.5 J$$

3. A circular coil having N turns and radius r carries a current I. It is held in the XZ plane in a magnetic field $B\hat{i}$. The torque on the coil due to the magnetic field is:

A. $B\pi r^2 IN$

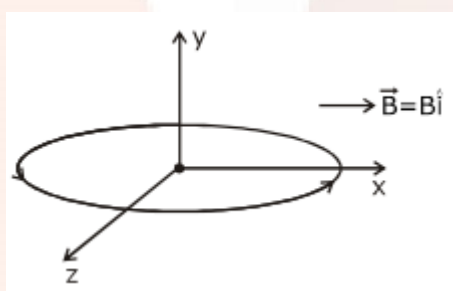
B. $\frac{Br^2 I}{\pi N}$

C. Zero

D. $\frac{B\pi r^2 I}{N}$

Ans. A

Explanation:



Let the current I is following in anti-clockwise direction, then the magnetic moment of the coil is

$$m = NIA$$

where, N = number of turns in coil

and A = area of each coil = πr^2 ,

The torque on the coil due to the magnetic field

$$\tau = MB \sin \theta$$

$$\tau = N I \pi r^2 B \sin 90^\circ = B \pi r^2 I N$$

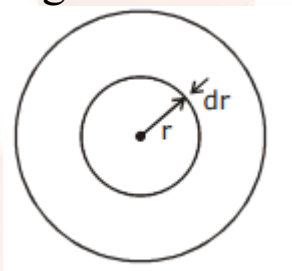
4. A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its center. The moment of Inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = aMR^2$. The value of the coefficient a is:

- A. $8/5$
- B. $1/2$
- C. $3/2$
- D. $3/5$

Ans. A

Explanation:

Consider an elementary ring of thickness dx and radius x .



$$M = \int_0^R (\rho_0 r) (2\pi r dr) = 2\pi \rho_0 \frac{R^3}{3}$$

$$I = MR^2 + \int_0^R 2\pi \rho_0 r^4 dr = MR^2 + \frac{2\pi \rho_0 R^5}{5}$$

$$I = MR^2 + \frac{3MR^2}{5} = \frac{8MR^2}{5}$$

5. Radiation coming from transitions $n=2$ to $n=1$ of hydrogen atoms fall on He^+ ions in $n=1$ and $n=2$ states. The possible transition of helium ions as they absorb energy from the radiation is:

- A. $n=2 \rightarrow n=4$
- B. $n=2 \rightarrow n=5$

C. $n=1 \rightarrow n=4$

D. $n=2 \rightarrow n=3$

Ans. A

Explanation:

For Hydrogen, Energy released

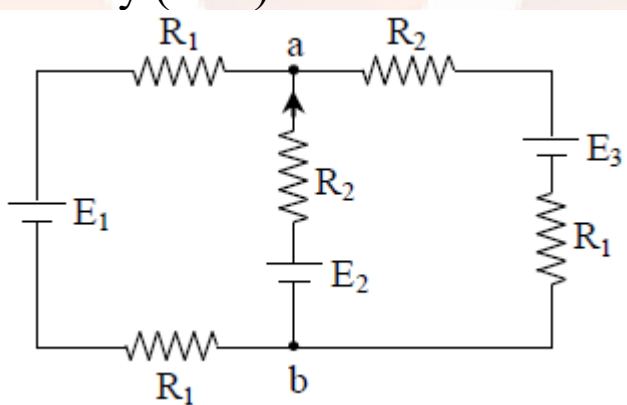
$$\Delta E = 13.6 \left[1 - \frac{1}{4} \right] = 10.2 \text{ eV}$$

for Helium, energy absorbed in Transition ($2 \rightarrow 4$)

$$\Delta E = 13.6 \left(\frac{1}{4} - \frac{1}{16} \right) 4 = 10.2 \text{ eV}$$

So, possible transition is $n = 2 \rightarrow n = 4$

6. For the circuit shown, with $R_1=1.0\Omega$, $R_2=2.0\Omega$, $E_1=2V$ and $E_2=E_3=4V$, the potential difference between the points 'a' and 'b' is approximately (in V) :



- A. 2.7 B. 2.3
C. 3.7 D. 3.3

Ans. D

Explanation:

Potential Difference the points 'a' and 'b' is

$$\Delta V = \frac{E_1 R_2 R_3 + E_2 R_1 R_3 + E_3 R_1 R_2}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

$$\Delta V = \frac{2(2 \times 2) + 4(2 \times 2) + 4(2 \times 2)}{4 + 4 + 4}$$

$$= \frac{40}{12} = \frac{10}{3} = 3.3 \text{ Volt}$$

7. If 10^{22} gas molecules each of mass 10^{-26} kg collide with a surface (perpendicular to it) elastically per second over an area 1 m^2 with a speed 10^4 m/s, the pressure exerted by the gas molecules will be of the order of :

A. 10^3 N/m^2 B. 10^{16} N/m^2
C. 10^8 N/m^2 D. 10^4 N/m^2

Ans. (bonus)

Explanation:

The pressure exerted by the gas molecules

$$P = \frac{(2mV)N}{\Delta t \times A} = \frac{2(10^{-26})(10^4)10^{22}}{1 \times 1} = 2 \frac{\text{N}}{\text{m}^2}$$

8. A solid conducting sphere, having a charge Q , is surrounded by an uncharged conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V . If the shell is now given a charge of $-4Q$, the new potential difference between the same two surfaces is:

A. 2 V B. -2 V
C. V D. 4 V

Ans. C

Explanation:

Electric fields between surfaces remain unchanged. So, shell will produce constant potential. Thus new potential difference = V

9. A steel wire having a radius of 2.0 mm, carrying a load of 4 kg, is hanging from a ceiling. Given that $g = 3.1\pi \text{ ms}^{-2}$, what will be the tensile stress that would be developed in the wire?
- A. $5.2 \times 10^6 \text{ Nm}^{-2}$
 B. $6.2 \times 10^6 \text{ Nm}^{-2}$
 C. $3.1 \times 10^6 \text{ Nm}^{-2}$
 D. $4.8 \times 10^6 \text{ Nm}^{-2}$

Ans. C

Explanation:

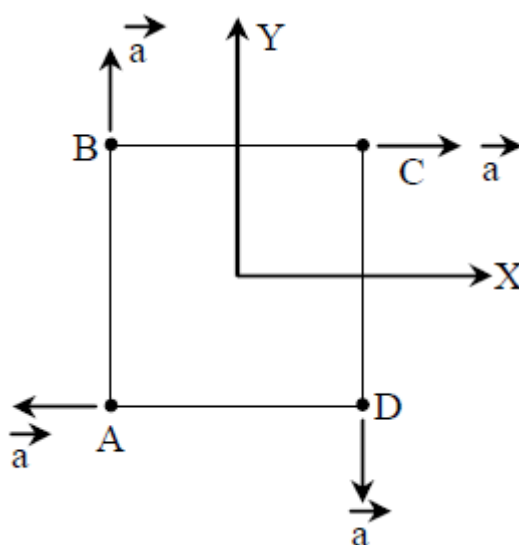
Given that radius of wire, $r = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$

Weight of load, $m = 4\text{kg}$, $g = 3.1 \pi \text{ ms}^{-2}$

$$\begin{aligned} \text{Tensile Stress} &= \frac{F}{A} = \frac{Mg}{\pi r^2} = \frac{4(3.1\pi)}{\pi 4 \times 10^{-6}} \\ &= 3.1 \times 10^6 \frac{\text{N}}{\text{m}^2} \end{aligned}$$

10. Four particles A, B, C and D with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4m$ are at the corners of a square. They have accelerations of the equal magnitude with directions as shown.

The acceleration of the centre of mass of the particles is :



A. $\frac{a}{5}(\hat{i} + \hat{j})$

B. Zero

C. $a(\hat{i} + \hat{j})$

D. $\frac{a}{5}(\hat{i} - \hat{j})$

Ans. C

Explanation

The acceleration of the centre of mass of the particles is

$$\vec{a}_{cm} = \frac{-ma\hat{i} + 2ma\hat{j} + 3ma\hat{i} - 4ma\hat{j}}{10m} = \frac{a}{5}[\hat{i} - \hat{j}]$$

11. A boy's catapult is made of rubber cord which is 42 cm long, with 6 mm diameter of cross-section and of negligible mass. The boy keeps a stone weighing 0.02 kg on it and stretches the cord by 20 cm by applying a constant force. When released, the stone flies off with a velocity of 20ms^{-1} . Neglect the change in the area of cross-section of the cord while stretched. The Young's modulus of rubber is closest to:

A. 10^6Nm^{-2}

B. 10^8Nm^{-2}

C. 10^4Nm^{-2}

D. 10^3Nm^{-2}

Ans. A

Explanation:

When rubber cord is stretched, then it stores potential energy and when released, this potential energy is given to the stone as kinetic energy.

So, potential energy of stretched cord = kinetic energy of stone

$$\frac{1}{2}Y\left(\frac{\Delta\ell}{\ell}\right)A\ell = \frac{1}{2}mV^2$$

$$\Rightarrow \text{Young's modulus of rubber, } Y = \frac{mV^2\ell}{A(\Delta\ell)^2}$$

$$Y = \frac{(0.02)(20)^2(0.42)}{\pi(9 \times 10^{-6})(0.2)^2} \approx 2.3 \times 10^6 \text{ N/m}^2$$

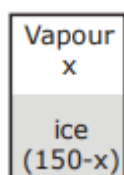
So, order is 10^6 .

12. A thermally insulated vessel contains 150 g of water at 0°C . Then the air from the vessel is pumped out adiabatically. A fraction of water turns into ice and the rest evaporates at 0°C itself. The mass of evaporated water will be closest to:
(Latent heat of vaporization of water = $2.10 \times 10^6 \text{ Jkg}^{-1}$ and Latent heat of Fusion of water = $3.36 \times 10^5 \text{ Jkg}^{-1}$)

- A. 150 g
- B. 130 g
- C. 35 g
- D. 20 g

Ans. D

Explanation:



Let amount of water evaporated be x gram

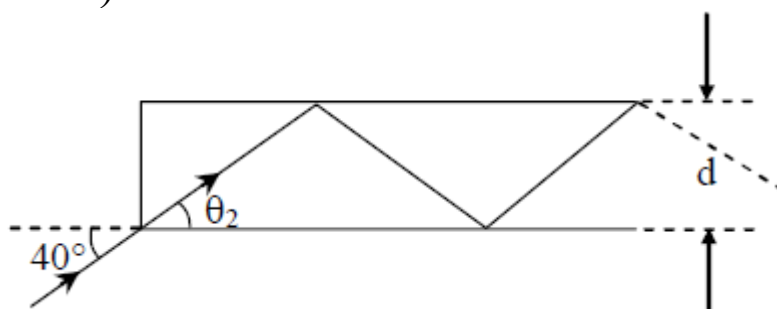
$$\therefore x \times L_v = (150 - x) \times L_s$$

$$x \times 540 = (150 - x) \times 80$$

$$\Rightarrow x = 20 \text{ g}$$

13. In figure, the optical fiber is $l=2$ m long and has a diameter of $d = 20 \mu\text{m}$. If a ray of light is incident on one end of the fiber at angle $\Theta_1=40^\circ$, the number of reflections it makes before emerging

from the other end is close to: (refractive index of fiber is 1.31 and $\sin 40^\circ = 0.64$)



- A. 66000 B. 57000
C. 45000 D. 55000

Ans. B

Explanation:

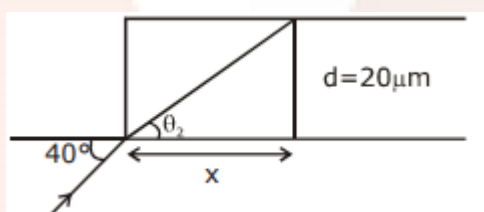
From Snell's Rule

$$\mu_1 \sin \theta_1 = \mu_2 \sin \theta_2$$

$$1 \sin 40 = 1.31 \sin \theta$$

$$\Rightarrow \sin \theta_2 = \frac{0.64}{1.31}$$

$$\theta_2 \approx 30^\circ$$



$$\tan 30^\circ = \frac{d}{x}$$

$$x = \frac{d}{\frac{1}{\sqrt{3}}} = 20\sqrt{3} \times 10^{-6} \text{ m}$$

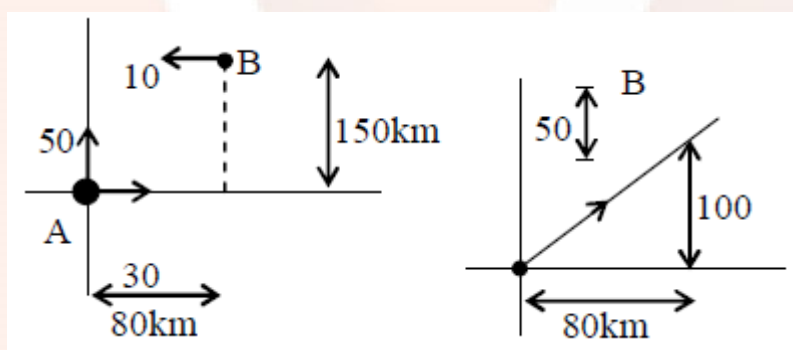
$$\text{No. of reflections} = \frac{20}{20\sqrt{3} \times 10^{-6}} \approx 57000$$

14. Ship A is sailing towards north-east with velocity $\vec{v} = 30\hat{i} + 50\hat{j}$ km/hr where $\hat{i}$ points east and $\hat{j}$ north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in :

- A. 3.2 hrs.
- B. 2.6 hrs.
- C. 4.2 hrs.
- D. 2.2 hrs.

Ans. B

Explanation:



Time after which the distance is minimum between A and B can be calculated as

$$t = \frac{|\vec{r}_{BA} \cdot \vec{V}_{AB}|}{|\vec{V}_{AB}|^2}$$

$$\vec{V}_{AB} = 40\hat{i} + 50\hat{j}$$

$$\vec{r}_{BA} = \vec{r}_B - \vec{r}_A = 80\hat{i} + 150\hat{j}$$

$$t_{\min} = \frac{|\vec{V}_{AB} \cdot \vec{r}_{BA}|}{|\vec{V}_{AB}|^2} = 2.6 \text{ hrs}$$

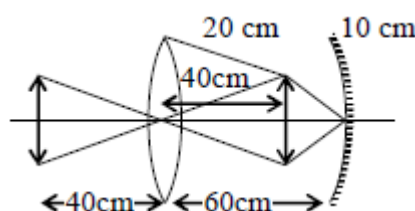
15. An upright object is placed at a distance of 40 cm in front of a convergent lens of focal length 20 cm. A convergent mirror of

focal length 10 cm is placed at a distance of 60 cm on the other side of the lens. The position and size of the final image will be:

- A. 20 cm from the convergent mirror, twice the size of the object.
- B. 40 cm from the convergent mirror, same size as the object.
- C. 40 cm from the convergent lens, twice the size of the object.
- D. 20 cm from the convergent mirror, same size as the object.

Ans. D

Explanation:

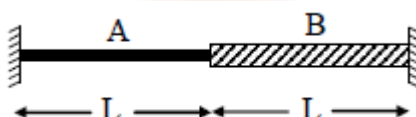


In given system of lens and mirror, position of object O in front of lens is at a distance $2f$.

i.e., $u = 2f = 40 \text{ cm}$

Image will form at object itself of same size and inverted.

16. A wire of length $2L$, is made by joining two wires A and B of same length but different radii r and $2r$ and made of the same material. It is vibrating at a frequency such that the joint of the two wires forms a node. If the number of antinodes in wire A is p and that in B is q then the ratio $p:q$ is:



- A. 3:5
- B. 1:2
- C. 4:9
- D. 1:4

Ans. B

Explanation:

Vibrating frequency is given by

$$f_1 = \frac{p}{2\ell} \sqrt{\frac{T}{\rho\pi r^2}} \text{ and } f_2 = \frac{q}{2\ell} \sqrt{\frac{T}{\rho 4\pi r^2}}$$

Now, $\therefore f_1 = f_2$

$$\text{So, } \frac{p}{q} = \frac{1}{2}$$

17. Water from a pipe is coming at a rate of 100 liters per minute. If the radius of the pipe is 5 cm, the Reynolds number for the flow is of the order of : (density of water = 1000 kg/m³, coefficient of Viscosity of water = 1 mPa s)

- A. 10⁶
- B. 10³
- C. 10⁴
- D. 10²

Ans. C

Explanation:

Flow rate of water (Q) = 100 lit/min

$$= \frac{100 \times 10^{-3}}{60} = \frac{5}{3} \times 10^{-3} \text{ m}^3$$

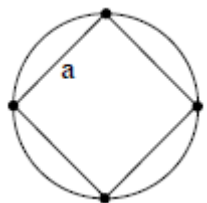
$$\text{Velocity of flow (v)} = \frac{Q}{A} = \frac{5 \times 10^{-3}}{3 \times \pi \times (5 \times 10^{-2})} = \frac{2}{3\pi} \text{ m/s}$$

$$R_e = \frac{dvp}{\eta}$$

$$= \frac{(10^{-1}) \left(\frac{2}{3\pi} \right) (1000)}{1} \approx 2.12 \times 10^4$$

$$\text{Order} = 10^4$$

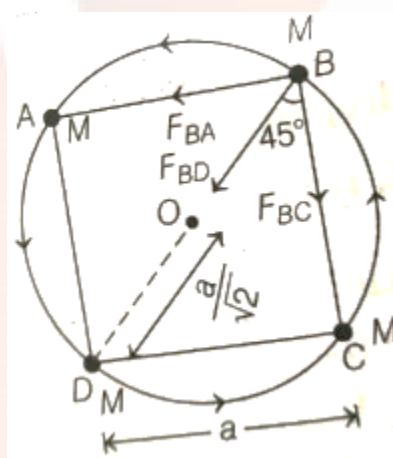
18. Four identical particles of mass M are located at the corners of a square of side ' a '. What should be their speed if each of them revolves under the influence of others' gravitational field in a circular orbit circumscribing the square?



- A. $1.16\sqrt{\frac{GM}{a}}$ B. $1.41\sqrt{\frac{GM}{a}}$
C. $1.21\sqrt{\frac{GM}{a}}$ D. $1.35\sqrt{\frac{GM}{a}}$

Ans. A

Explanation:



Net force on mass M at position B towards center circle is

$$\begin{aligned} F_{BO_{\text{net}}} &= F_{BD} + F_{BA} \sin 45^\circ + F_{BC} \cos 45^\circ \\ &= \frac{GM^2}{(\sqrt{2}a)^2} + \frac{GM^2}{a^2} \left(\frac{2}{\sqrt{2}} \right) = \frac{GM^2}{a^2} \left(\frac{1}{2} + \sqrt{2} \right) \end{aligned}$$

From above figure,

$$r = \frac{a}{\sqrt{2}}, \quad F = \frac{GM^2}{a^2}(\sqrt{2}) + \frac{GM^2}{2a^2}$$

$$\therefore \frac{GM^2}{a^2} \left(\sqrt{2} + \frac{1}{2} \right) = \frac{MV^2}{a/\sqrt{2}}$$

$$\Rightarrow V = \sqrt{\frac{GM}{a}} \sqrt{1 + \frac{1}{2\sqrt{2}}} = 1.16 \sqrt{\frac{GM}{a}}$$

19. The wavelength of the carrier waves in a modern optical fiber communication network is close to:

- A. 2400 nm B. 600 nm
C. 900 nm D. 1500 nm

Ans. D

Explanation:

$\lambda \approx 1550$ nm (most widely used wavelength in optical fiber system)

20. The bob of a simple pendulum has mass 2 g and a charge of $5.0 \mu\text{C}$. It is at rest in a uniform horizontal electric field of intensity 2000 V/m . At equilibrium, the angle that the pendulum makes with the vertical is:

(take $g = 10 \text{ m/s}^2$)

- A. $\tan^{-1}(0.5)$
B. $\tan^{-1}(2.0)$
C. $\tan^{-1}(5.0)$
D. $\tan^{-1}(0.2)$

Ans. A

Explanation:

For the equilibrium

$$T \cos \theta = mg$$

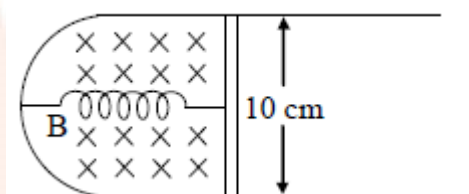
$$\text{and } T \sin \theta = qE$$

At equilibrium, the angle that the pendulum makes with the vertical is given by

$$\tan \theta = \frac{qE}{Mg} = \frac{5 \times 10^{-6} \times 2 \times 10^3}{2 \times 10^{-3} \times 10} = \frac{1}{2}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

21. A thin strip 10 cm long is on a U shaped wire of negligible resistance and it is connected to a spring of spring constant 0.5 Nm^{-1} (see figure). The assembly is kept in a uniform magnetic field of 0.1 T. If the strip is pulled from its equilibrium position and released, the number of oscillations it performs before its amplitude decreases by a factor of e is N . If the mass of the strip is 50 grams, its resistance 10Ω and air drag negligible, N will be close to :



- A. 1000
- B. 10000
- C. 50000
- D. 5000

Ans. D

Explanation:

Spring force = kx

where, k = spring constant

As the slider is kept in a uniform magnetic field $B = 0.1 \text{ T}$, hence it will experience a force, i.e.

Magnetic force = Bil

where, l = length of the strip,

Now using

$F_{\text{net}} = ma$

$$F = -kx - ilB = -kx - \frac{vIB}{R}$$

$$-KX - \frac{v\ell^2 B^2}{R} = ma$$

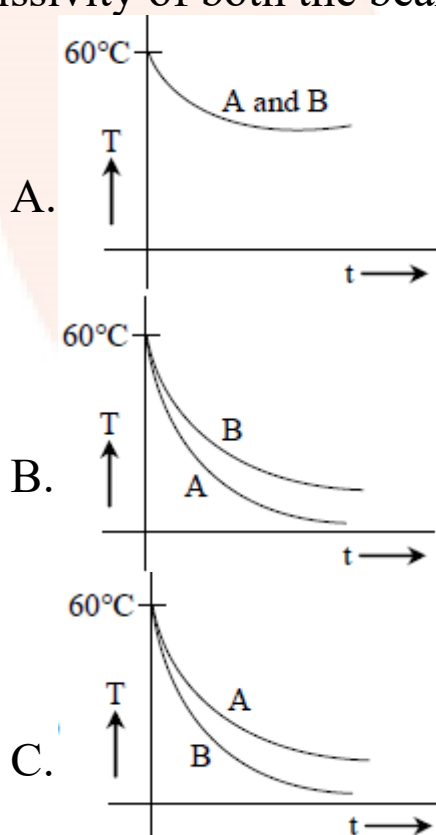
$$A = A_0 e^{-bt/em}$$

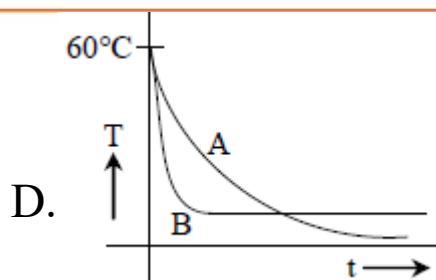
$$t = \frac{2mR}{B^2 \ell^2} = \frac{2(50 \times 10^{-3})(10)}{(0.1)^2(0.1)^2} = 10^4$$

$$t = 2\pi\sqrt{m/K} = 2\text{ sec} \Rightarrow f = 0.5\text{ Hz}$$

$$N = 5000$$

22. Two identical beakers A and B contain equal volumes of two different liquids at 60°C each and left to cool down. Liquid in A has density of $8 \times 10^2 \text{ kg/m}^3$ and specific heat of $2000 \text{ J kg}^{-1} \text{ K}^{-1}$ while Liquid in B has density of 10^3 kg m^{-3} and specific heat of $4000 \text{ J kg}^{-1} \text{ K}^{-1}$. Which of the following best describes their temperature versus time graph schematically? (Assume the emissivity of both the beakers to be the same)





Ans. B

Explanation:

From the Newton's law of cooling, we have rate of cooling.

$$\frac{dQ}{dt} = \frac{h}{ms}(T - T_0)$$

where, h = heat transfer coefficient

T = temperature of body.

T_0 = temperature of surrounding

m = mass and s = specific heat.

We know, $m = V \cdot \rho$

where, V = volume and ρ = density

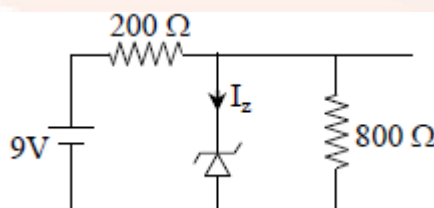
$$-\frac{dT}{dt} = \frac{e\sigma A}{ms} 4T_0^3 (T - T_0) \propto \frac{1}{\rho S}$$

$$\text{For } (\rho s)_A = 800 \times 2000 = 16 \times 10^5$$

$$(\rho s)_B = 10^3 \times 4000 = 40 \times 10^5$$

$$(\rho s)_B > (\rho s)_A \Rightarrow \left(-\frac{dT}{dt}\right)_B < \left(-\frac{dT}{dt}\right)_A$$

23. The reverse breakdown voltage of a Zener diode is 5.6 V in the given circuit.



The current I_Z through the Zener is:

- A. 10 mA
- B. 7 mA
- C. 17 mA
- D. 15 mA

Ans. A

Explanation:

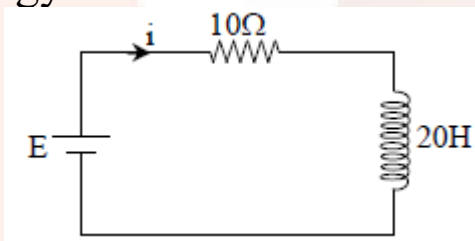
where, potential drop across $800\ \Omega$ resistance = potential drop across Zener diode = 5.6 V

$$I_{200\ \Omega} = \frac{9 - 5.6}{200} = \frac{3.4}{200}\text{ mA}$$

$$I_{800\ \Omega} = \frac{5.6}{800}\text{ mA}$$

$$I_z = \frac{3.4}{200} - \frac{5.6}{800} = 10\text{ mA}$$

24. A 20 Henry inductor coil is connected to a 10 ohm resistance in series as shown in figure. The time at which rate of dissipation of energy (joule's heat) across resistance is equal to the rate at which magnetic energy is stored in the inductor, is :

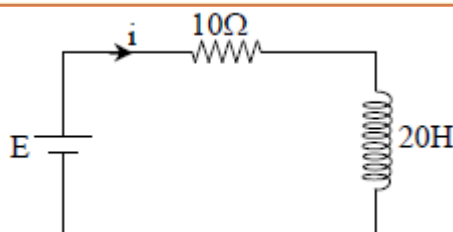


- A. $\frac{1}{2} \ln 2$
- B. $2 \ln 2$
- C. $\ln 2$
- D. $\frac{2}{\ln 2}$

Ans. B

Explanation:

Given circuit is a series L. R circuit



In an L-R circuit, current increases as

$$i = \frac{E}{R} \left(1 - e^{-\frac{R}{L}t} \right)$$

Now, energy stored in inductor is

$$U_L = \frac{1}{2} Li^2$$

where, L = self-inductance of the coil and energy dissipated by resistor is

$$U_R = i^2 R$$

Given, rate of energy stored in inductor is equal to the rate of energy dissipation in resistor. So, after differentiating, we get

$$i^2 R = \left(\tau \frac{di}{dt} \right) i$$

$$\Rightarrow iR = \tau \frac{di}{dt} \Rightarrow \frac{di}{dt} = \frac{i}{\tau}$$

$$e^{-t/\tau} = 1 - e^{-t/\tau}$$

$$\Rightarrow 2e^{-t/\tau} = 1 \Rightarrow t = \tau \ln 2$$

$$\tau = \frac{L}{R} = \frac{20}{10} = 2$$

$$t = 2 \ln 2$$

25. An alternating voltage $v(t) = 220 \sin 100\pi t$ volt is applied to a purely resistive load of 50Ω . The time taken for the current to rise from half of the peak value to the peak value is :

A. 2.2 ms

B. 5 ms

C. 3.3 ms

D. 7.2 ms

Ans. C

Explanation:

In an AC resistive circuit current and voltage are in phase

$$\text{so } I = \frac{V}{R} \Rightarrow I = \frac{220}{50} \sin(100\pi t) \quad \dots(i)$$

$$\Delta\phi = \frac{\pi}{3} = (100\pi)\Delta t$$

The time duration in which current reaches half of its maximum value after reaching maximum value is

$$\Rightarrow \Delta t = \frac{10^3}{300} \text{ms} = 3.3 \text{ms}$$

26. Two particles move at right angle to each other. Their de Broglie wavelengths are λ_1 and λ_2 respectively. The particles suffer perfectly inelastic collision. The de Broglie wavelength λ , of the final particle, is given by :

A. $\lambda = \frac{\lambda_1 + \lambda_2}{2}$

B. $\frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$

C. $\lambda = \sqrt{\lambda_1 \lambda_2}$

D. $\frac{2}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

Ans. D

Explanation:

Given, de-Broglie wavelengths for particles are λ_1 and λ_2 .

$$\text{So, } \lambda_1 = \frac{h}{p_1} \text{ and } \lambda_2 = \frac{h}{p_2}$$

The de Broglie wavelength λ , of the final particle is given by

$$p_1^2 + p_2^2 = p^2$$

$$\Rightarrow \frac{1}{\lambda^2} = \frac{1}{\lambda_1^2} + \frac{1}{\lambda_2^2}$$

27. Voltage rating of a parallel plate capacitor is 500 V. Its dielectric can withstand a maximum electric field of 10^6 m^2 . The plate area is 10^{-4} m^2 . What is the dielectric constant if the capacitance is 15 pF ?(given $\epsilon_0 = 8.86 \times 10^{-12} \text{ C}^2/\text{Nm}^2$)

- A. 4.5 B. 3.8
C. 6.2 D. 8.5

Ans. D

Explanation:

As we know, capacitance of a capacitor filed with dielectric medium.

$$C = \frac{A \epsilon_0 K}{d} \Rightarrow C = \frac{A \epsilon_0 KE}{V} \quad (\because V = Ed)$$

$$K = \frac{CV}{A \epsilon_0 E} = \frac{(15 \times 10^{-12})(500)}{10^{-4}(8.86 \times 10^{-12})(10^6)} \\ = 8.5$$

28. In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is:

- A. $\text{A}^{-1} \text{TML}^3$ B. $\text{A}^2 \text{T}^3 \text{M}^{-1} \text{L}^{-2}$
C. $\text{AT}^2 \text{M}^{-1} \text{L}^{-1}$ D. $\text{AT}^{-3} \text{ML}^{3/2}$

Ans. B

Explanation:

The dimensions of $\epsilon_0 = \text{M}^{-1} \text{A}^2 \text{L}^{-3} \text{T}^4$

The dimensions of $\mu_0 = \text{MLT}^{-2} \text{A}^{-2}$

So, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}} = [\text{M}^{-2} \text{A}^4 \text{L}^{-4} \text{T}^6]^{1/2} = \text{M}^{-1} \text{A}^2 \text{L}^{-2} \text{T}^3$

29. In an interference experiment the ratio of amplitudes of coherent waves is $\frac{a_1}{a_2} = \frac{1}{3}$. The ratio of maximum and minimum intensities of fringes will be:

- A. 2 B. 4
C. 9 D. 18

Ans. B

Explanation:

In Young's double slit experiment, ratio of maxima and minima intensity is given by \

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2$$

As, intensity (I) $\propto$ [amplitude (a)²]

It is given that

$$a_1 : a_2 = 1 : 3$$

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2 = \left(\frac{4}{2} \right)^2 = 4 : 1$$

30. A plane electromagnetic wave travels in free space along the x- direction. The electric field component of the wave at a particular point of space and time is $E = 6 \text{ Vm}^{-1}$ along y- direction. Its corresponding magnetic field component, B would be:

- A. $6 \times 10^{-8} \text{ T}$ along z- direction
B. $2 \times 10^{-8} \text{ T}$ along z- direction
C. $2 \times 10^{-8} \text{ T}$ along y- direction
D. $6 \times 10^{-8} \text{ T}$ along x- direction

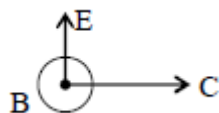
Ans. B

Explanation:

For an electromagnetic wave, ratio of magnitudes of electric and magnetic field is

$$\frac{E}{B} = c.$$

Where, c is the speed electromagnetic wave in vacuum.



$$B = \frac{E}{c} = \frac{6}{3 \times 10^8} = 2 \times 10^{-8} (+z)$$

Chemistry

1. The size of the iso-electronic species Cl^- , Ar and Ca^{2+} is affected by :

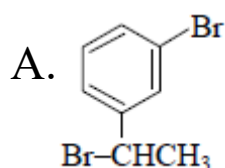
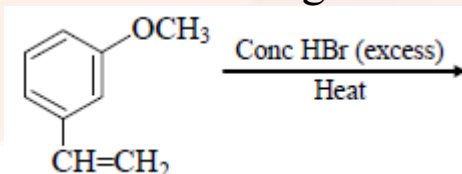
- A. nuclear charge
- B. azimuthal quantum number of valence shell
- C. electron –electron interaction in the outer orbitals
- D. Principle quantum number of valence shell

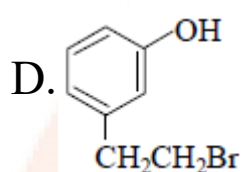
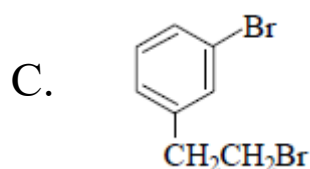
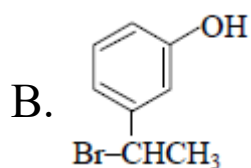
Ans. A

Explanation:

Iso-electronic species differ in size due to different effective nuclear charge.

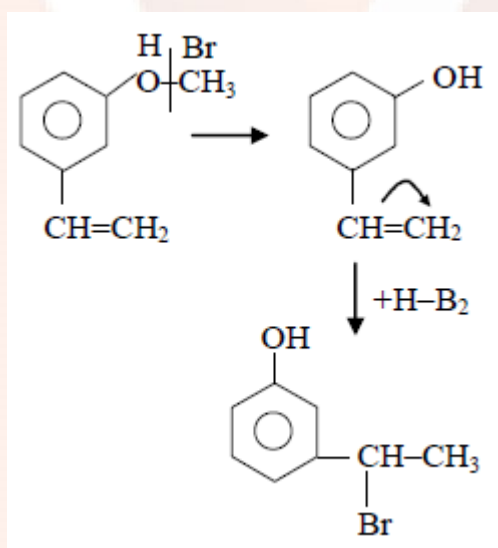
2. The major product of the following reaction is :





Ans. B

Explanation:



3. Element 'B' forms ccp structure and 'A' occupies half of the octahedral voids, while oxygen atoms occupy all the tetrahedral voids. The structure of bimetallic oxide is :

- A. A_2BO_4 B. A_2B_2O
C. A_4B_2O D. AB_2O_4

Ans. B

Explanation:

Lattice formed by B (ccp) = 4

A = 50% of octahedral voids = 2

O = tetrahedral voids = 8

Formula = AB₂O₄

4. Given that = $E_{O_2/H_2O}^\ominus = +1.23V$; $E_{S_2O_8^{2-}/SO_4^{2-}}^\ominus = 2.05V$

$E_{Br_2/Br^-}^\ominus = +1.09V$; $E_{Au^{3+}/Au}^\ominus = +1.4V$

The strongest oxidizing agent is -

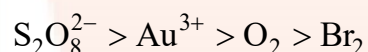
- A. S₂O₈²⁻ B. O₂
C. Br₂ D. Au³⁺

Ans. A

Explanation:

Strongest oxidising agent $\propto$ lowest position in electrochemical series (ECS)

The decreasing order of oxidizing agent among the given option is as follows:



5. Which is wrong with respect to our responsibility as a human being to protect our environment?

- A. Setting up compost tin in gardens.
B. Restricting the use of vehicles
C. Avoiding the use of floodlighted facilities
D. Using plastic bags.

Ans. D

Explanation:

Use of plastic bags is hazardous to our environment.

6. Maltose on treatment with dilute HCl gives:

- A. D-Fructose
- B. D—Glucose and D—Fructose
- C. D-Galactose
- D. D-Glucose

Ans. D

Explanation:

Hydrolysis of maltose give glucose as maltose is composed of two α -D glucose units.

7. Which of the following amines can be prepared by Gabriel phthalimide reaction?

- A. Triethylamine
- B. t-butylamine
- C. neo-pentylamine
- D. n-butylamine

Ans. D

Explanation:

n-butylamine ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$) can be prepared by Gabriel phthalimide reaction.

This method produces only primary amines without the traces of secondary or tertiary amines.

n – Butyl amine



only unhindered 1° Amines are products of this reaction.

8. The vapour pressures of pure liquids A and B are 400 and 600 mmHg, respective at 298 K. On mixing the two liquids, the sum of their initial volumes is equal to the volume of the final mixture. The molefraction of liquid B is 0.5 in the mixture. The vapour pressure of the final solution, the mole fractions of components A and B in vapour phase, respectively are :
- A. 450 mmHg, 0.5, 0.5
B. 500 mmHg, 0.4, 0.6
C. 450 mmHg, 0.4, 0.6
D. 500 mmHg, 0.5, 0.5

Ans. D

Explanation:

The vapour pressures of pure liquids A , $P_A^\circ = 400 \text{ mmHg}$

The vapour pressures of pure liquids B, $P_B^\circ = 600 \text{ mmHg}$

$x_B = 0.5$, $P_S = ?$ $y_A = ?$, $y_B = ?$

$$P_S = P_A^\circ x_A + P_B^\circ x_B$$

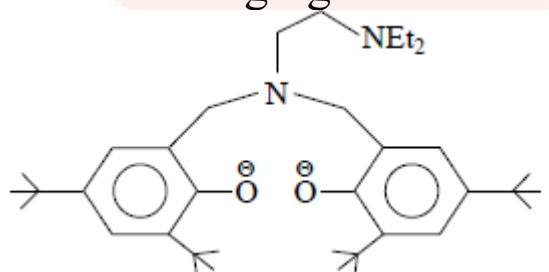
$$= 400(0.5) + 600(0.5)$$

$$= 200 + 300 = 500$$

$$y_A = \frac{P_A}{P_S} = \frac{200}{500} = \frac{2}{5} = 0.4$$

$$y_B = \frac{P_B}{P_S} = \frac{300}{500} = 0.6$$

9. The following ligand is :



- A. tetradentate
B. bidentate
C. hexadentate

D. tridentate

Ans. A

Explanation:

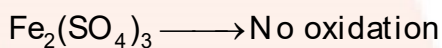
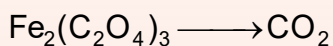
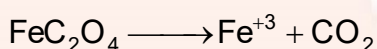
Tetradentate ligands have been widely used in coordination, biological, synthetic, and supramolecular chemistry. It has four lone pairs but maximum it will be able to donate three lone pairs.

10. In order to oxidise a mixture of one mole of each of FeC_2O_4 , $\text{Fe}(\text{C}_2\text{O}_4)_3$, FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ in acidic medium, the number of moles of KMnO_4 required is:

- A. 2 B. 1.5
C. 1 D. 3

Ans. A

Explanation:



$$g_m E(\text{KMnO}_4)$$

$$= g_m E(\text{FeC}_2\text{O}_4) + G_m E[\text{Fe}_2(\text{C}_2\text{O}_4)_3]$$

$$+ g_m E(\text{FeSO}_4) \text{ moles} \times V.F.$$

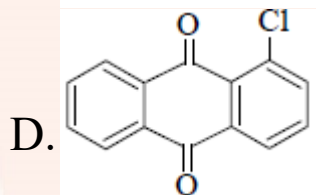
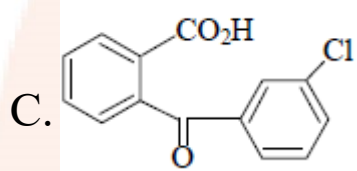
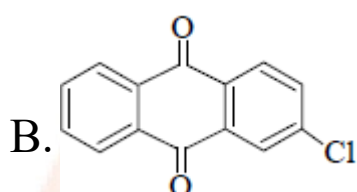
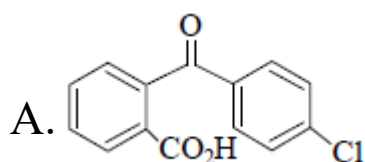
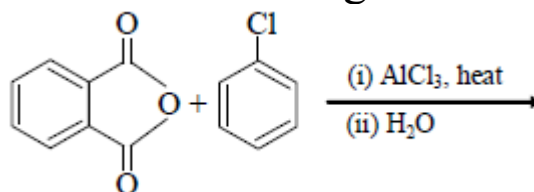
$$= \text{moles} \times V.F. + \text{moles} \times V.F. + \text{mole} \times V.F.$$

$$x \times 5 = (1 \times 3) + (1 \times 6) + (1 \times 1)$$

$$5x = 10 \quad (\text{MnO}_4^- \longrightarrow \text{Mn}^{+2})$$

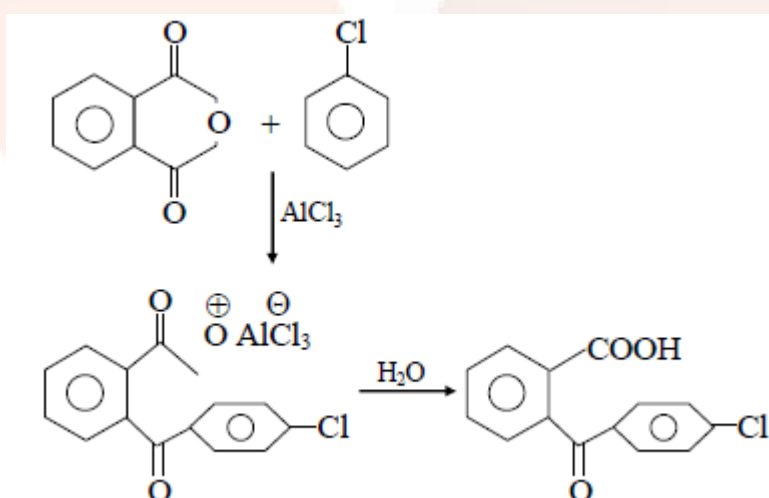
$$x = 2$$

11. The major product of the following reaction is :



Ans. A

Explanation:



12. The correction order of hydration enthalpies of alkali metal ions is :

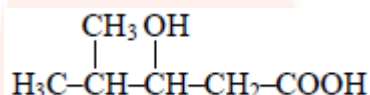
- A. $\text{Na}^+ > \text{Li}^+ > \text{K}^+ > \text{Cs}^+ > \text{Rb}^+$
- B. $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$
- C. $\text{Na}^+ > \text{Li}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$
- D. $\text{Li}^+ > \text{Na}^+ > \text{K}^+ > \text{Cs}^+ > \text{Rb}^+$

Ans. B

Explanation:

Smaller is size more is hydration energy. So, Li^+ having minimum radius so maximum hydration and maximum hydration radii.

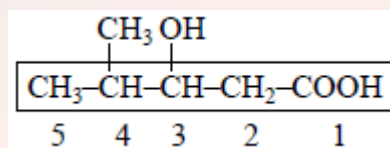
13. The IUPAC name of the following compound is :



- A. 3—Hydroxy-4—methylpentanoic acid
- B. 4—Methyl-3-hydroxypentanoic acid
- C. 2—Methyl-3-hydroxypentan-5-oic acid
- D. 4,4-Dimethyl-3-hydroxybutanoic Acid

Ans.A

Explanation:



The IUPAC name of the compound is 3—Hydroxy-4—methylpentanoic acid

14. The lanthanide ion that would Show colour is :

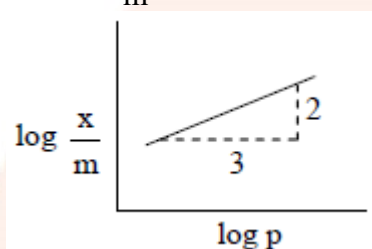
- A. La^{3+}
- B. Sm^{3+}
- C. Gd^{3+}
- D. Lu^{3+}

Ans. B

Explanation:

Sm^{+3} show yellow color due to presence of unpaired e^- .

15. Adsorption of a gas follows Freundlich adsorption isotherm. x is the mass of the gas adsorbed on mass m of the adsorbent. The plot $\log \frac{x}{m}$ versus $\log p$ is shown in the given graph. $\frac{x}{m}$ is proportional to :



- A. $p^{3/2}$ B. p^3
C. p^2 D. $p^{2/3}$

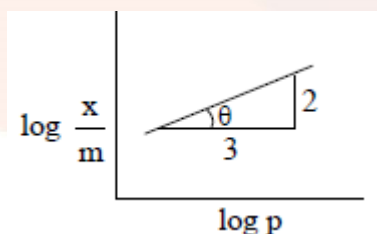
Ans. D

Explanation:

According to Freundlich,

$$\frac{x}{m} = kp^{1/n} [n > 1]$$

where, m = mass of adsorbent, x = mass of the gas adsorbed,
 x/m = amount of gas adsorbed per unit mass of solid adsorbent,
 p = pressure, K and n = constants.



$$\frac{x}{m} = Kp^{1/n}$$

$$\log \frac{x}{m} = \frac{1}{n} \log p + \log K$$

$$\text{Slope} = \tan \theta = \frac{2}{3} = \frac{1}{n}$$

$$\frac{x}{m} \propto p^{2/3}$$

16. For the reaction $2A + B \rightarrow C$, the values of initial rate at different reactant concentrations are given in the table below. The rate law for the reaction is :

[A](mol L ⁻¹)	[B](mol L ⁻¹)	Initial Rate (mol L ⁻¹ s ⁻¹)
0.05	0.05	0.045
0.10	0.05	0.090
0.20	0.10	0.72

- A. Rate = $k[A]^2[B]$
- B. Rate = $k[A][B]^2$
- C. Rate = $k[A]^2[B]^2$
- D. Rate = $k[A][B]$

Ans. B

Explanation:

Rate is given by

$$r = k[A]^p[B]^q$$

$$\frac{r_2}{r_1} = \left[\frac{A_2}{A_1} \right]^p \left[\frac{B_2}{B_1} \right]^q$$

$$2^1 = 2^p \quad (p = 1)$$

$$\frac{r_3}{r_2} = \left[\frac{A_3}{A_2} \right]^1 \left[\frac{B_3}{B_2} \right]^q$$

$$\frac{0.720}{0.090} = 2^q = 4 = 2^2$$

$$q = 2$$

$$r = k[A]^1[B]^2$$

17. The correct order of the spin-only magnetic moment of metal ions in the following low—spin complexes, $[\text{V}(\text{CN})_6]^{4-}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Ru}(\text{NH}_3)_6]^{3+}$, and $[\text{Cr}(\text{NH}_3)_6]^{2+}$, is :

- A. $\text{Cr}^{2+} > \text{Ru}^{3+} > \text{Fe}^{2+} > \text{V}^{2+}$
- B. $\text{V}^{2+} > \text{Cr}^{2+} > \text{Ru}^{3+} > \text{Fe}^{2+}$
- C. $\text{V}^{2+} > \text{Ru}^{3+} > \text{Cr}^{2+} > \text{Fe}^{2+}$
- D. $\text{Cr}^{2+} > \text{V}^{2+} > \text{Ru}^{3+} > \text{Fe}^{2+}$

Ans. B

Explanation:

	No. of unpaired electron
$[\text{V}(\text{CN})_6]^{4-} \text{V}^{2+}$	3
$[\text{Fe}(\text{CN})_6]^{4-} \text{Fe}^{2+}$	0
$[\text{Ru}(\text{NH}_3)_6]^{3+} \text{Ru}^{3+}$	1
$[\text{Cr}(\text{NH}_3)_6]^{2+} \text{Cr}^{2+}$	2

$\therefore$ Order of spin magnetic moment



18. The quantum number of four electrons are given below :

- I. $n = 4, l = 2, m_l = -2, m_s = -1/2$
- II. $n = 3, l = 2, m_l = 1, m_s = +1/2$
- III. $n = 4, l = 1, m_l = 0, m_s = +1/2$
- IV. $n = 3, l = 1, m_l = 1, m_s = -1/2$

The correct order of their increasing energies will be :

- A. I < II < III < IV
- B. I < III < II < IV

C. $IV < III < II < I$

D. $IV < II < III < I$

Ans. D

Explanation:

Smaller the value of $(n + l)$, smaller the energy. If two or more sub-orbits have same values of $(n + l)$, sub-orbits with lower values of n has lower energy. The $(n + l)$ values of the given options are as follows:

I. $n + l = 6$

II. $n + l = 5$

III. $n + l = 5$

IV. $n + l = 4$

The correct order of their increasing energies will be

$IV < II < III < I$

19. In following compounds, the decreasing order of the basic strength will be :

A. $NH_3 > C_2H_5NH_2 > (C_2H_5)_2NH$

B. $(C_2H_5)_2NH > C_2H_5NH_2 > NH_3$

C. $(C_2H_5)_2NH > NH_3 > C_2H_5NH_2$

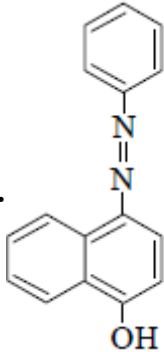
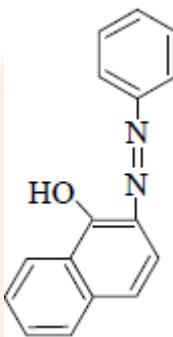
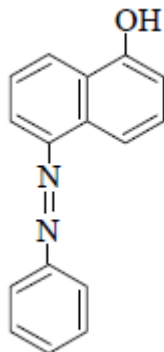
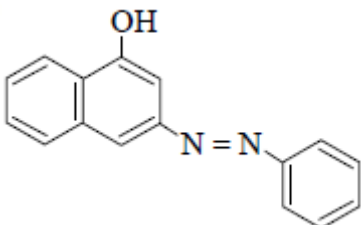
D. $C_2H_5NH_2 > NH_3 > (C_2H_5)_2NH$

Ans. B

Explanation:

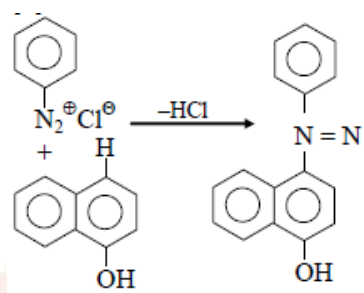
In aqueous medium sec. amines are most basic. 3° amines are more basic than NH_3 as +I factor dominate over steric factor.

20. Coupling of benzene diazonium chloride with 1-naphthol in alkaline medium will give :

- A. 
- B. 
- C. 
- D. 

Ans. A

Explanation:



21. Which of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas ? (Assume non—expansion work is zero)

- A. Adiabatic process : $\Delta U = -w$
- B. Isochoric process : $\Delta U = q$
- C. Cyclic process : $q = -w$
- D. Isothermal process : $q = -w$

Ans. D

Explanation:

From the 1st law of thermodynamics

$$\Delta U = q + W$$

where, ΔU = change in internal energy

q = heat

W = work done

1. Adiabatic process

$$q = 0, (\Delta E = w) \text{ or } (\Delta U = w)$$

2. Cyclic process

$$\Delta E = 0, (q = -w)$$

3. Isothermal process

$$\Delta E = nCV\Delta T, (\Delta T = 0)$$

$$\Delta E = 0, (q = -w)$$

4. Isochoric process $\Delta V = 0$

$$w = P\Delta V = 0$$

$$\Delta E = q$$

22. An organic compound 'X' showing the following solubility profile is :

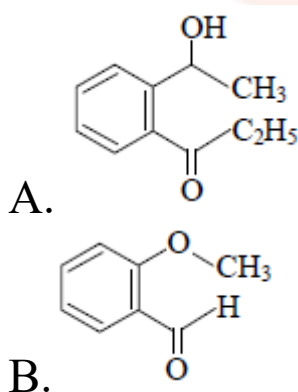
'X'	Water	→ insoluble
	5% HCl	→ insoluble
	10% NaOH	→ soluble
	10% NaHCO ₃	→ insoluble

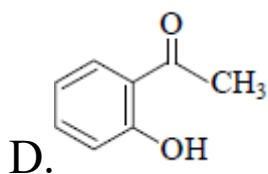
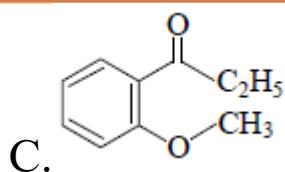
- A. Oleic acid
- B. O-Toluidine
- C. M-Cresol
- D. Benzamide

Ans. C

Both m-(re so) and olek and form salt with 10% NaOH, but m=(re so) salt is soluble whereas oeic acid salt is insoluble due to very long unsaturated carbon chain

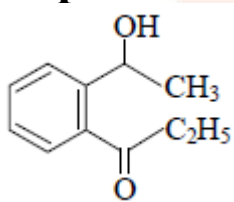
23. An organic compound neither reacts with neutral ferric chloride solution nor with Fehling solution. It however, reacts with Grignard reagent and gives positive iodoform tests The compound is :





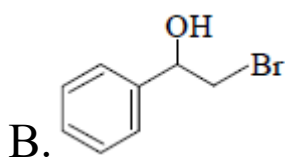
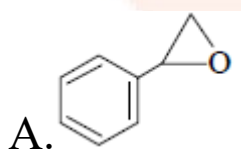
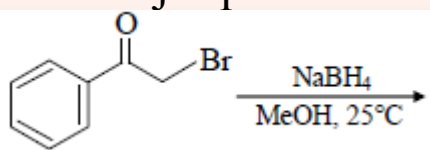
Ans. A

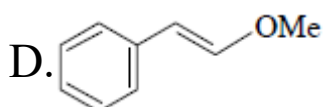
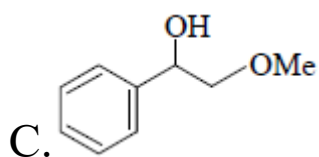
Explanation:



- Reacts with Grignard's reagent due to acidic hydrogen.
- Fehling solution test is negative as there is no -CHO group.
- Neutral FeCl_3 test is negative as there is no phenolic group

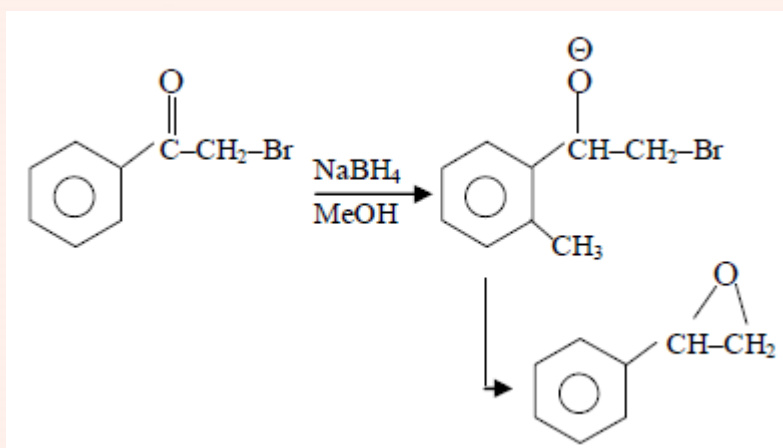
24. The major product of the following reaction is :





Ans. A

Explanation:



25. With respect to an ore, Ellingham diagram helps to predict the feasibility of its

- A. Thermal reduction
- B. Vapour phase refining
- C. Zone refining
- D. Electrolysis

Ans. A

Explanation:

Ellingham diagram is used to select reducing agent so it help to predict feasibility of its thermal reduction.

26. Diborane (B_2H_6) reacts independently with O_2 and H_2O to produce, respectively :

- A. B_2O_3 and $[BH_4]^-$
- B. HBO_2 and H_3BO_3
- C. B_2O_3 and H_3BO_3
- D. H_3BO_3 and B_2O_3

Ans. C

Explanation:

Diborane (B_2H_6) reacts independently with O_2 and H_2O to produce B_2O_3 and H_3BO_3 respectively.



27. For silver $C_p(JK^{-1} mol^{-1}) = 23 + 0.01T$. the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH will be close to:

- A. 62 kJ B. 16 kJ
- C. 21 kJ D. 13 kJ

Ans. A

Explanation:

It is given that

$$C_p = 23 + 0.01 T$$

$$n = 3 \text{ moles}$$

$$T_1 = 300 \text{ K}$$

$$T_2 = 1000 \text{ K}$$

$$\Delta H = nC_p \Delta T$$

According to Kirchoff's relation

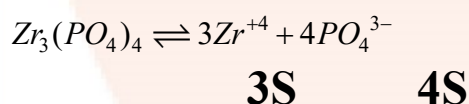
$$\begin{aligned}\Delta H &= n \int_{T_1}^{T_2} C_p \Delta T \\ &= 3 \int_{300}^{1000} (23 + 0.01T) dT \\ &= 3 \int_{300}^{1000} \left(23T + \frac{0.01T^2}{2} \right) \\ &= 61.95 \text{ kJ/mole} \\ &= 62 \text{ kJ/mole}\end{aligned}$$

28. If solubility product of $Zr_3(PO_4)_4$ is denoted by K_{sp} and its molar solubility is denoted by S , then which of the following relation between S and K_{sp} is correct ?

A. $S = \left(\frac{K_{sp}}{144} \right)^{1/6}$ B. $S = \left(\frac{K_{sp}}{929} \right)^{1/9}$
C. $S = \left(\frac{K_{sp}}{6912} \right)^{1/7}$ D. $S = \left(\frac{K_{sp}}{216} \right)^{1/7}$

Ans. C

Explanation:



$$K_{sp} = (3S)^3 (4S)^4$$

$$K_{sp} = 6912S^7$$

$$S = \left(\frac{K_{sp}}{6912} \right)^{1/7}$$

29. Assertion : Ozone is destroyed by CFCs in the upper stratosphere,

Reason: Ozone holes increase the amount of UV radiation reaching the earth.

A. Assertion and reason are incorrect

Assertion and reason are both correct,

B. and the reason is the correct explanation for the assertion.

C. Assertion and reason are correct, but the reason is not the explanation for the assertion.

D. Assertion is false, but the reason is correct.

Ans. C

Explanation:

CFC's are responsible for depletion of ozone layer.

30. 100 mL of a water sample contains 0.81 g of calcium bicarbonate and 0.73 g of magnesium bicarbonate. The hardness of this water sample expressed in terms of equivalents of CaCO_3 is :

(Molar mass of calcium bicarbonate is 162 g mol^{-1} and magnesium bicarbonate is 146 g mol^{-1})

A. 100 ppm B. 5,000 ppm

C. 1,000 ppm D. 10,000 ppm

Ans. D

Explanation:

Given, $W_{\text{Ca}(\text{HCO}_3)_2} = 0.81 \text{ g}$

$W_{\text{Mg}(\text{HCO}_3)_2} = 0.73 \text{ g}$

$M_{\text{Ca}(\text{HCO}_3)_2} = 162 \text{ g mol}^{-1}$

$M_{\text{Mg}(\text{HCO}_3)_2} = 146 \text{ g mol}^{-1}$

$V_{\text{H}_2\text{O}} = 100 \text{ mL}$

ppm of $\text{CaCO}_3 = ?$

degree of hardness = $\frac{\text{weight of hardness causing salt}}{\text{Mw}} \times 100$

1 ppm = 1 part CaCO_3 eq in 10^6 parts water

$$\begin{aligned}\text{ppm of CaCO}_3 &= \frac{\left(\frac{0.73}{146} + \frac{0.81}{162}\right) \times 100}{100} \times 10^6 \\ &= 10^4 \text{ ppm} \\ &= 10,000 \text{ ppm}\end{aligned}$$

MATHEMATICS

1. The area (in sq units) of the region
 $A = \{(x, y) \in \mathbf{R} \times \mathbf{R} \mid 0 \leq x \leq 3, 0 \leq y \leq 4, y \leq x^2 + 3x\}$ is :
- A. $53/6$ B. 8
 C. $59/6$ D. $26/3$

Ans. C

Explanation:

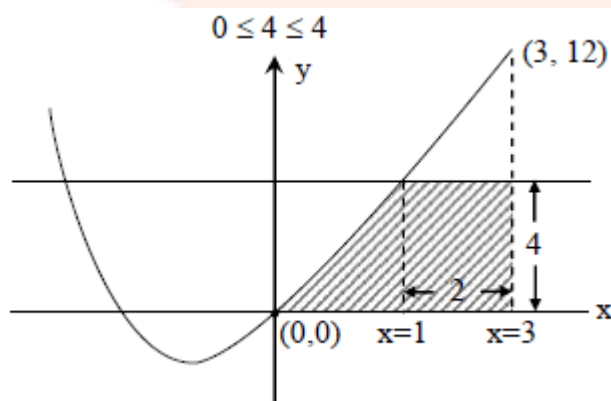
Given, $y \leq x^2 + 3x$

$$\Rightarrow y \leq \left(x + \frac{3}{2}\right)^2 - \frac{9}{4}$$

$$\Rightarrow \left(x + \frac{3}{2}\right)^2 \geq \left(y + \frac{9}{4}\right)$$

Since, $0 \leq y \leq 4$ and $0 \leq x \leq 3$

$\therefore$ The diagram for the given inequalities is



Required region

$$\begin{aligned}
 &= \int_0^1 [x^2 + 3x] dx + [2 \times 4] \\
 &= \left[\frac{x^3}{3} + \frac{3}{2}x^2 \right]_0^1 + 8 \\
 &= \frac{2+9+48}{6} = \frac{59}{6}
 \end{aligned}$$

2. Let $f:[0,2] \rightarrow \mathbf{R}$ be a twice differentiable function such that $f''(x) > 0$, for all $x \in (0,2)$. If $\Phi(x) = f(x) + (2-x)$, then Φ is
- Increasing on $(0, 2)$
 - Increasing on $(0, 1)$ and decreasing on $(1,2)$.
 - Decreasing on $(0,2)$
 - Decreasing on $(0,1)$ and increasing on $(1,2)$

Ans. D

Explanation:

Given that

$$\phi(x) = f(x) + f(2-x), \forall x \in (0,2)$$

$$\Rightarrow \phi'(x) = f'(x) - f'(2-x)$$

Also, we have $f''(x) > 0 \forall x \in (0,2)$

$\downarrow$	$\downarrow$
$\phi(x)$ is increasing i.e. $\phi'(x) > 0$ $f'(x) - f'(2-x) > 0$ $f'(x) > f'(2-x)$ $x > 2-x$ $2x > 2$ $x > 1$ i.e. $x \in (1, 2)$	$\phi(x)$ is decreasing i.e. $\phi'(x) < 0$ $f'(x) - f'(2-x) < 0$ $f'(x) < f'(2-x)$ $x < 2-x$ $x > 1$ i.e. $x \in (0, 1)$

3. The greatest value of $c \in \mathbf{R}$ for which the system of linear equations

$$x - cy - cz = 0$$

$$cx - y + cz = 0$$

$$cx + cy - z = 0$$

has a non-trivial solution, is :

A. 2

B. $\frac{1}{2}$

C. 0

D. -1

Ans. B

Explanation:

If the system of equations has non-trivial solutions, then

$$\begin{vmatrix} 1 & -c & -c \\ c & -1 & c \\ c & c & -1 \end{vmatrix} = 0$$

$$1(1 - c^2) + c(-c - c^2) - c(c^2 + c) = 0$$

$$1 - c^2 - c^2 - c^3 - c^3 - c^2 = 0$$

$$2c^3 + 3c^2 - 1 = 0$$

$$c = -1$$

$$c = \frac{1}{2}$$

Maximum value of $c = \frac{1}{2}$

4. The equation of a plane containing the line of intersection of the planes $2x - y - 4 = 0$ and $y + 2z - 4 = 0$ and passing through the point $(1, 1, 0)$ is:

A. $2x - z = 2$

B. $x + 3y + z = 4$

C. $x - 3y - 2z = -2$

D. $x - y - z = 0$

Ans. D

Explanation:

Equation of given planes are

$$2x - y - 4 = 0 \quad \dots(i)$$

$$\text{and } y + 2z - 4 = 0 \quad \dots(ii)$$

Now, equation of family of planes passes through the line of intersection of given planes (i) and (ii) is

$$(2x - y - 4) + \lambda (y + 2z - 4) = 0$$

Required plane passes through (1, 1, 0)

$$(2 - 1 - 4) + \lambda(1 + 0 - 4) = 0$$

$$\lambda = -1$$

$$(2x - y - 4) - (y + 2z - 4) = 0$$

$$2x - 2y - 2z = 0$$

$$x - y - z = 0$$

5. If $\cos(\alpha + \beta) = \frac{3}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then $\tan(2\alpha)$ is equal to:

A. $\frac{21}{16}$

B. $\frac{33}{52}$

C. $\frac{63}{16}$

D. $\frac{63}{52}$

Ans. C

Explanation:

It is given that

$$\cos(\alpha + \beta) = \frac{3}{5} \quad \Rightarrow \quad \tan(\alpha + \beta) = \frac{4}{3}$$

$$\sin(\alpha - \beta) = \frac{5}{13} \quad \Rightarrow \quad \tan(\alpha - \beta) = \frac{5}{12}$$

Now,

$$\tan(2\alpha) = \tan((\alpha + \beta) + (\alpha - \beta))$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta)\tan(\alpha - \beta)}$$

$$= \frac{\frac{4}{3} + \frac{5}{12}}{1 - \frac{4}{3} \times \frac{5}{12}} = \frac{48 + 15}{36 - 20} = \frac{63}{16}$$

6. Let $y = y(x)$ be the solution of the differential equation, $(x^2 + 1)^2 \frac{dy}{dx} + 2x(x^2 + 1)y = 1$ such that $y(0) = 0$. If $\sqrt{a} y(1) = \frac{\pi}{32}$, then the value of 'a' is :

A. $\frac{1}{16}$

B. 1

C. $\frac{1}{4}$

D. $\frac{1}{2}$

Ans. A

Explanation:

Given differential equation is

$$(x^2 + 1)^2 \frac{dy}{dx} + 2x(x^2 + 1)y = 1$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{2x}{1+x^2} \right) y = \frac{1}{(1+x^2)^2}$$

[Dividing each term by $(1+x^2)^2$] .. (i)

This is linear differential equation of the form $\frac{dy}{dx} + P.y = Q$

Here, $P = \frac{2x}{(1+x^2)}$ and $Q = \frac{1}{(1+x^2)^2}$

$$\text{I.F.} = e^{\int \frac{2x}{1+x^2} dx} = (1+x^2)$$

Solution of diff. eq.

$$y(1+x^2) \times \int \frac{1}{(1+x^2)^2} \times (1+x^2) dx$$

$$y(1+x^2) = \tan^{-1}(x) + c$$

Given that $y(0) = 0$

$$c = 0$$

$$y(1+x^2) = \tan^{-1}(x)$$

Put $x = 1$

$$y \cdot (2) = \frac{\pi}{4}$$

$$y = \frac{\pi}{8}$$

$$\frac{1}{4} \cdot y = \frac{\pi}{32}$$

$$\text{i.e., } \sqrt{a} = \frac{1}{4}$$

$$a = \frac{1}{16}$$

7. All possible numbers are formed using the digits 1,1,2,2,2,2,3,4,4 taken all at a time. The number of such numbers in which the odd digits occupy even places is :

- A. 175 B. 162
C. 180 D. 160

Ans. C

Explanation:

There are total 9 digits; out of which only 3 digits are odd.

$$\text{Number of ways to arrange odd digits first} = \left({}^4C_3 \times \frac{3!}{2!} \right)$$

Total number of possible numbers

$$= \left({}^4C_3 \times \frac{3!}{2!} \right) \left(\frac{6!}{4!2!} \right) \\ = 180$$

8. If

$$2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2, x \in \left(0, \frac{\pi}{2} \right)$$

then $\frac{dy}{dx}$ is equal to :

- A. $x - \frac{\pi}{2}$ B. $\frac{\pi}{6} - x$
C. $2x - \frac{\pi}{3}$ D. $\frac{\pi}{3} - x$

Ans. B

Explanation:

$$2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2 \quad x \in \left(0, \frac{\pi}{2} \right)$$

[dividing each term of numerator and denominator by $\cos x$]

$$2y = \left(\cot^{-1} \left(\frac{\sqrt{3} + \tan x}{1 - \sqrt{3} \tan x} \right) \right)^2$$

$$2y = \left(\frac{\pi}{2} - \tan^{-1} \left(\tan \left(\frac{\pi}{3} + x \right) \right) \right)^2$$

$$0 < x < \frac{\pi}{6}$$

$$\left(\frac{\pi}{3} < x + \frac{\pi}{3} < \frac{\pi}{2} \right)$$

$$2y = \left(\frac{\pi}{2} - \frac{\pi}{3} - x \right)^2$$

$$2y = \left(\frac{\pi}{6} - x \right)^2$$

Differentiation

$$2 \cdot \frac{dy}{dx} = 2 \left(\frac{\pi}{6} - x \right) (-1)$$

$$\frac{dy}{dx} = x - \frac{\pi}{6}$$

$$\text{at } x = \frac{\pi}{6}$$

$f \times u$ is undefined
(non-diff)

$$\frac{\pi}{6} < x < \frac{\pi}{2}$$

$$\frac{\pi}{2} < x + \frac{\pi}{3} < \frac{5\pi}{6}$$

$$2y = \left(\frac{\pi}{2} - \left(\frac{\pi}{3} + x - \pi \right) \right)^2$$

$$2y = \left(\frac{2\pi}{6} - x \right)^2$$

Differentiation

$$2 \cdot \frac{dy}{dx} = 2 \left(\frac{7\pi}{6} - x \right) (-1)$$

$$\frac{dy}{dx} = x - \frac{7\pi}{6}$$

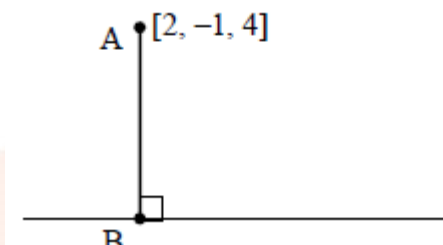
9. The length of the perpendicular from the point $(2, -1, 4)$ on the straight line,

$$\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1} \text{ is:}$$

- A. greater than 4
- B. less than 2
- C. greater than 2 but less than 3
- D. greater than 3 but less than 4

Ans. D

Explanation:



Let B be the foot of perpendicular from point A(2, -1, 4) on the given line.

So, B can be assumed as

$$[10\lambda - 3, -7\lambda + 2, \lambda] \quad \frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1} = \lambda$$

$$\text{Drs of AB} = (10\lambda - 5, -7\lambda + 3, \lambda - 4)$$

$$\text{Drs of given line} = (10, -7, 1)$$

$$10(10\lambda - 5) - 7(-7\lambda + 3) + (\lambda - 4) = 0$$

$$100\lambda - 50 + 49\lambda - 21 + \lambda - 4 = 0$$

$$150\lambda = 75$$

$$\lambda = \frac{1}{2}$$

$$\text{Point B} \left(2, -\frac{3}{2}, \frac{1}{2} \right)$$

$$\text{Distance length AB} = \frac{5}{\sqrt{2}} = 3.53$$

10. The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n$, $n \in \mathbb{N}$, where $\mathbb{N}$ is the set of all natural number is:

- A. 160 B. 320
C. 105 D. 210

Ans. D

Explanation:

Equation of circle $x^2 + y^2 = 16$

Equation of chord $x + y = n$ $n \in \mathbb{N}$ where $\mathbb{N}$ is the set of all natural number

Length of chord $= 2\sqrt{16 - \frac{n^2}{2}}$

$(\text{Length of chord})^2 = 2(32 - n^2)$

Possible value of $n = 1, 2, 3, 4, 5$

Sum of squares of the length of chords
 $= 2(31) + 2(28) + 2(23) + 2(16) + 2(7)$
 $= 210$

11. $\int \frac{\sin \frac{5x}{2}}{\sin \frac{x}{2}} dx$ is equal to :

(Where, c is a constant of integration.)

- A. $2x + \sin x + \sin 2x + c$
 B. $2x + \sin x + 2 \sin 2x + c$
 C. $x + 2 \sin x + \sin 2x + c$
 D. $x + 2 \sin x + 2 \sin 2x + c$

Ans. C

Explanation:

It is given that

$$I = \int \frac{\sin\left(\frac{5x}{2}\right)}{\sin\left(\frac{x}{2}\right)} dx$$

[Multiplying by $\cos\frac{x}{2}$ in numerator and denominator]

$$I = \int \frac{2\sin\left(\frac{5x}{2}\right)\cos\left(\frac{x}{2}\right)}{\sin x} dx$$

$$I = \int \frac{\sin(3x) + \sin(2x)}{\sin x} dx$$

$$I = \int \frac{(3\sin x - 4\sin^3 x) + 2\sin x \cos x}{(\sin x)} dx$$

$$I = \int (3 - 4\sin^2 x + 2\cos x) dx$$

$$I = \int (3 - 2(1 - \cos 2x) + 2\cos x) dx$$

$$I = \int (3 - 2 + 2\cos 2x + 2\cos x) dx$$

$$I = \int (1 + 2\cos 2x + 2\cos x) dx$$

$$I = x + \sin 2x + 2 \sin x + c$$

12. The magnitude of the projection of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ on the vector perpendicular to the plane containing the vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$, is :

A. $\frac{\sqrt{3}}{2}$

B. $\sqrt{6}$

C. $\frac{\sqrt{3}}{2}$

D. $3\sqrt{6}$

Ans.

A

Explanation:

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

Vector $\perp$ to given to vectors is

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

$$\vec{a} \times \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$$

Projection of vector $\vec{c} = 2\hat{i} + 3\hat{j} + \hat{k}$ on $\vec{a} \times \vec{b} = \frac{|\vec{c} \cdot (\vec{a} \times \vec{b})|}{|\vec{a} \times \vec{b}|}$

$$= \frac{|(\hat{i} - 2\hat{j} + \hat{k}) \cdot (2\hat{i} + 3\hat{j} + \hat{k})|}{\sqrt{6}}$$

$$= \frac{|2 - 6 + 1|}{\sqrt{6}} = \frac{-3}{\sqrt{6}} = \sqrt{\frac{3}{2}}$$

13. If the tangents on the ellipse $4x^2 + y^2 = 8$ at the points (1,2) and (a,b) are perpendicular to each other, then a^2 is equal to :

A. $\frac{2}{17}$

B. $\frac{64}{17}$

C. $\frac{4}{17}$

D. $\frac{128}{17}$

Ans.

A

Explanation:

The Equation of ellipse is given as $4x^2 + y^2 = 8 \dots(i)$

$$\frac{dy}{dx} = -\frac{4x}{y}$$

And, it is also given that the tangent at (1, 2) and (a, b) are perpendicular to each other,

$$\left(-\frac{y}{2}\right)\left(-\frac{4a}{b}\right) = -1$$

So, solving the above equation, we get,

$$b = -8a \dots(i)$$

Since, (a, b) lies on ellipse

Thus,

$$4a^2 + b^2 = 8 \text{ (from eq. (i))}$$

$$4a^2 + 64a^2 = 8$$

$$a^2 = \frac{8}{68} = \frac{2}{17}$$

14. The sum of the series $2 \cdot {}^{20}C_0 + 5 \cdot {}^{20}C_1 + 8 \cdot {}^{20}C_2 + 11 \cdot {}^{20}C_3 + \dots + 62 \cdot {}^{20}C_{20}$ is equal to :

- A. 2^{24} B. 2^{26}
C. 2^{23} D. 2^{25}

Ans. D

Explanation:

It is given that

$$S = 2 \cdot {}^{20}C_0 + 5 \cdot {}^{20}C_1 + \dots + 59 \cdot {}^{20}C_{19} + 62 \cdot {}^{20}C_{20} \dots \text{(i)}$$

$$S = 62 \cdot {}^{20}C_{20} + 59 \cdot {}^{20}C_{19} + \dots + 5 \cdot {}^{20}C_1 + 2 \cdot {}^{20}C_0 \dots \text{(ii)}$$

Now, adding equation (i) & (ii)

$$2S = 64 [{}^{20}C_0 + {}^{20}C_1 + {}^{20}C_2 + \dots + {}^{20}C_{20}]$$

$$2S = 64 \cdot 2^{20}$$

$$S = 64 \cdot 2^{19}$$

$$S = 2^6 \cdot 2^{19} = 2^{25}$$

Thus, The sum of the series $2 \cdot {}^{20}C_0 + 5 \cdot {}^{20}C_1 + 8 \cdot {}^{20}C_2 + 11 \cdot {}^{20}C_3 + \dots + 62 \cdot {}^{20}C_{20}$ is equal to 2^{25} .

15. Let A and B be two non-null event such that $A \subset B$. Then which of the following statement is always correct?

- A. $P(A|B) = 1$
B. $P(A|B) \leq P(A)$
C. $P(A|B) = P(B) - P(A)$
D. $P(A|B) \geq P(A)$

Ans. D

Explanation:

It is given that

$$\because A \subset B, \therefore A \cap B = A$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}$$

$P(B)$ lies between $(0, 1]$

So,

$$P(A/B) \geq P(A)$$

16. The contrapositive of the statement “If you are born in India, then you are a citizen of India”, is :

- A. If you are not born in India, then you are not a citizen of India.
- B. If you are not a citizen of India, then you are not born in India.
- C. If you are born in India, then you are not a citizen of India.
- D. If you are a citizen of India, then you are born in India.

Ans. B

Explanation:

Contrapositive of given statement is. If you are not a citizen of India, then you are not born in India.

17. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, (\alpha \in \mathbb{R})$ such that

$$A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \text{ Then a value of } \alpha \text{ is:}$$

- A. $\frac{\pi}{32}$
- B. 0
- C. $\frac{\pi}{64}$
- D. $\frac{\pi}{16}$

Ans. C

Explanation:

It is given that matrix

$$A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

Similarly, we get that,

$$A^{32} = \begin{bmatrix} \cos 32\alpha & -\sin 32\alpha \\ \sin 32\alpha & \cos 32\alpha \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$32\alpha = \frac{\pi}{2}$$

$$\alpha = \frac{\pi}{64}$$

Thus, the value of α is $\frac{\pi}{64}$.

18. A point on the straight line, $3x + 5y = 15$ Which is equidistant from the coordinate axes will lie only in :

- A. 1st and 2nd quadrants
- B. 1st, 2nd and 4th quadrants
- C. 1st quadrant
- D. 4th quadrant

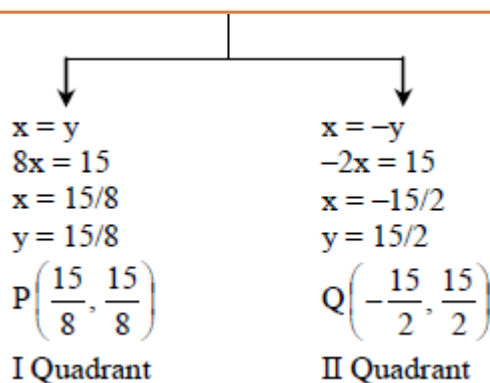
Ans. A

Explanation:

Given equation of line is $3x + 5y = 15$... (i)

Clearly, a point on the line (i) which is equidistance from X and Y-axes will lie on the line either $y = x$ or $y = -x$.

$3x + 5y = 15$ equation of straight line



19. The sum of all natural 'n' such that $100 < n < 200$ and H.C.F. $(91, n) > 1$ is :

- A. 3221 B. 3121
C. 3303 D. 3203

Ans. B

Explanation:

The natural numbers between 100 and 200 are 101, 102, 103, ..., 199

Since, $91 = 13 \times 7$, so the natural numbers between 100 and 200 whose HCF with 91 is more than 1 are the numbers which are either divisible by 7 or 13.

So, the required sum of numbers between 100 and 200 = (sum of numbers divisible by 7) + (sum of numbers divisible by 13) – (sum of numbers divisible by 91).

H.C.F. $(91, n) > 1$

Where, $100 < n < 200$

Sum of all positive values of 'n'

$$= 7 [15 + 16 + \dots + 28] + 13 [8 + \dots + 15] - (13 \times 14)$$

$$= 7 \times \frac{14}{2} [15 + 28] + 13 \times \frac{8}{2} [8 + 15] - (13 \times 14)$$

$$= (49 \times 43) + (52 \times 23) - 182$$

$$= 2107 + 1196 - 182$$

$$= 3121$$

20. The sum of the co-efficients of all even degree terms in 'x' in the expansion of $(x + \sqrt{x^3 - 1})^6 + (x - \sqrt{x^3 - 1})^6, (x > 1)$ is equal to :

- A. 29 B. 32
C. 26 D. 24

Ans. D

Explanation:

Use formula:

$$(a + b)^n + (a - b)^n = 2[nC_0 a^{n+2n} C_2 a^{n-2} b^2 + nC_4 a^{n-4} b^4 + \dots]$$

The given expression is

$$(x + \sqrt{x^3 - 1})^6 + (x - \sqrt{x^3 - 1})^6$$

$$= 2 [{}^6C_0 x^6 (x^3 - 1)^0 + {}^6C_2 x^4 (x^3 - 1) + {}^6C_4 x^2 (x^3 - 1)^2 + {}^6C_6 x^0 (x^3 - 1)^3]$$

Sum of coefficient at all even power

$$= 2 [{}^6C_0 + {}^6C_2 \{-1\} + {}^6C_4 x^2 \{1 + 1\} + {}^6C_6 \{-{}^3C_1 - {}^3C_3\}]$$

$$= 2 [1 + 15(-1) + 15(2) + (-4)]$$

$$= 2 [1 - 15 + 30 - 4]$$

$$= 2 \times 12 = 24$$

21. Let O (0,0) and A (0,1) be two fixed points. Then the locus of a point P such that the perimeter of ΔAOP is 4, is :

- A. $9x^2 - 8y^2 + 8y = 16$
B. $9x^2 + 8y^2 - 8y = 16$
C. $8x^2 + 9y^2 - 9y = 18$
D. $8x^2 - 9y^2 + 9y = 18$

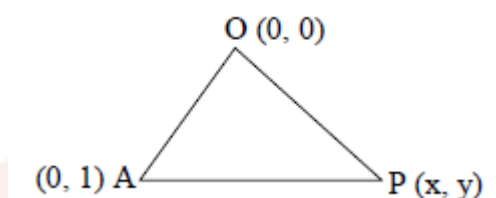
Ans. B

Explanation:

Given vertices of ΔAOP are $O(0,0)$ and $A(0, 1)$

Let the coordinates of point P are (x, y) .

In Triangle AOP



Given that

$$OA + AP + OP = 4$$

$$1 + \sqrt{x^2 + (y-1)^2} + \sqrt{x^2 + y^2} = 4$$

$$\left(\sqrt{x^2 + y^2}\right)^2 = \left(3 - \sqrt{x^2 + (y-1)^2}\right)^2$$

$$x^2 + y^2 = 9 + x^2 + y^2 - 2y + 1 - 6\sqrt{x^2 + (y-1)^2}$$

$$3\sqrt{x^2 + (y-1)^2} = (5 - y)$$

$$9(x^2 + (y-1)^2) = (5 - y)^2$$

$$9x^2 + 8y^2 - 8y = 16$$

22. If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is :

- | | |
|------|------|
| A. 2 | B. 3 |
| C. 5 | D. 4 |

Ans. D

Explanation:

Given Equation:

$$x^2 - 2x + 2 = 0$$

$$(x - 1)^2 = -1$$

$$x = 1 + i \text{ \& } 1 - i$$

$$\text{Let } \alpha = 1 + i$$

$$\beta = 1 - i$$

$$\frac{\alpha}{\beta} = \frac{1+i}{1-i} = \frac{(1+i)^2}{2} = \frac{1+i^2+2i}{2}$$

$$\frac{\alpha}{\beta} = i$$

$$\left(\frac{\alpha}{\beta}\right)^n = (i)^n = 1$$

Least value of $n = 4$

23. If S_1 and S_2 are respectively the sets of local minimum and local maximum points of the function, $f(x) = 9x^4 + 12x^3 - 36x^2 + 25, x \in \mathbf{R}$, then :

A. $S_1 = \{-2\}; S_2 = \{0, 1\}$

B. $S_1 = \{-2, 1\}; S_2 = \{0\}$

C. $S_1 = \{-2, 0\}; S_2 = \{1\}$

D. $S_1 = \{-1\}; S_2 = \{0, 2\}$

Ans. B

Explanation:

Given function:

$$f(x) = 9x^4 + 12x^3 - 36x^2 + 25$$

$$f'(x) = 36x^3 + 36x^2 - 72x$$

$$f'(x) = 36x(x^2 + x - 2)$$

$$f'(x) = 36x(x+2)(x-1)$$

Sign of $f'(x)$

⊖	⊕	⊖	⊕
-2	0	1	
↑	↑	↑	
Minima	Maxima	Minima	

$$S_1 = \{-2, 1\}$$

$$S_2 = \{0\}$$

24. The sum of the solutions of the equation $|\sqrt{x}-2|+\sqrt{x}(\sqrt{x}-4)+2=0, (x > 0)$ equal to :

- A. 10 B. 9
C. 4 D. 12

Ans. A

Explanation:

Given equation is

$$|\sqrt{x}-2|+\sqrt{x}(\sqrt{x}-4)+2=0 \quad (x > 0)$$

$$\Rightarrow |\sqrt{x}-2|+x-4\sqrt{x}+4=2$$

$$\Rightarrow |\sqrt{x}-2|+(\sqrt{x}-2)^2=2$$

$$\Rightarrow (|\sqrt{x}-2|)^2+|\sqrt{x}-2|-2=0$$

$$\text{let } \sqrt{x}=t$$

$$|t-2|+t(t-4)+2=0$$

Now, we have two cases.

Case I:

$$t \geq 2$$

$$t-2+t^2-4t+2=0$$

$$t^2-3t=0$$

$$t=0(\text{Reject})$$

$$t=3$$

$$\sqrt{x}=3$$

$$x=9$$

Case II:

$$t < 2$$

$$2 - t + t^2 - 4t + 2 = 0$$

$$t^2 - 5t + 4 = 0$$

$$t = 4(\text{Reject})$$

$$t = 1$$

$$\sqrt{x} = 1$$

$$x = 1$$

So, sum of all solution $= 9 + 1 = 10$

25. The shortest distance between the line $y = x$ and the curve $y^2 = x - 2$ is :

A. $\frac{7}{8}$

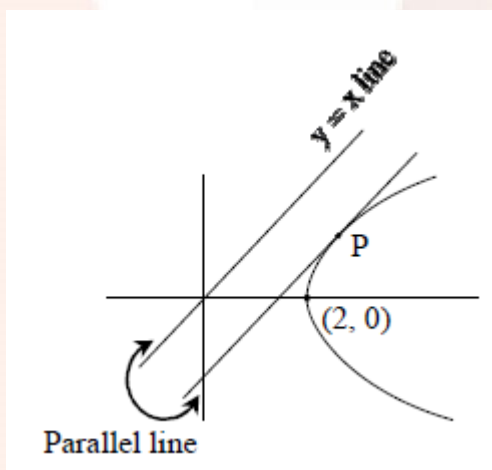
B. 2

C. $\frac{11}{4\sqrt{2}}$

D. $\frac{7}{4\sqrt{2}}$

Ans. D

Explanation:



$y = x$ line

$y^2 = x - 2$ parabola

for point P $\left(\frac{dy}{dx}\right)_p = 1$

$$\left(\frac{1}{2y}\right)_p = 1$$

$$(y_{\text{coordinate}}) = \frac{1}{2}$$

$$(x_{\text{coordinate}}) = \frac{9}{4}$$

Point P $\left(\frac{9}{4}, \frac{1}{2}\right)$

Shortest distance = length of perpendicular from P on line $x - y = 0$

$$= \frac{\left|\frac{9}{4} - \frac{1}{2}\right|}{\sqrt{2}} = \frac{7}{4\sqrt{2}}$$

26 If $f(x) = \frac{2 - x \cos x}{2 + x \cos x}$ and $g(x) = \log_e x$ ($x > 0$) then the value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g(f(x)) dx$ is :

A. $\log_e 3$

B. $\log_e 2$

C. $\log_e 1$

D. $\log_e e$

Ans. C

Explanation:

Given functions are

$$f(x) = \frac{2 - x \cos x}{2 + x \cos x} \quad g(x) = \log_e x (x > 0)$$

$$I = \int_{-\pi/4}^{\pi/4} f(f(x)) dx$$

$$I = \int_{-\pi/4}^{\pi/4} \log_e \left(\frac{2 - x \cos x}{2 + x \cos x} \right) dx \quad \dots(i)$$

Applying,

$$I = \int_{-\pi/4}^{\pi/4} \log_e \left(\frac{2 + x \cos x}{2 - x \cos x} \right) dx \quad \dots(ii)$$

So, now adding equation (i) and (ii), we get,

$$2I = \int_{-\pi/4}^{\pi/4} \log_e(1) dx$$

$$\Rightarrow 2I = 0$$

$$\Rightarrow I = 0$$

27. If $\alpha = \cos^{-1}\left(\frac{3}{5}\right)$, $\beta = \tan^{-1}\left(\frac{1}{3}\right)$, where $0 < \alpha, \beta < \frac{\pi}{2}$, then $\alpha - \beta$ is

equal to:

A. $\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

B. $\cos^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

C. $\tan^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

D. $\tan^{-1}\left(\frac{9}{14}\right)$

Ans. A

Explanation:

Given that

$$\alpha = \cos^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\beta = \tan^{-1}\left(\frac{1}{3}\right)$$

Now, we have to find the value of $\alpha - \beta$

$$\alpha - \beta = \tan^{-1}\left(\frac{4}{3}\right) - \tan^{-1}\left(\frac{1}{3}\right)$$

$$\left[\because \tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}, \text{ if } xy > -1 \right]$$

$$\begin{aligned}
 &= \tan^{-1} \left(\frac{\frac{4}{3} - \frac{1}{3}}{1 + \frac{4}{3} \times \frac{1}{3}} \right) \\
 &= \tan^{-1} \left(\frac{1}{1 + \frac{4}{9}} \right) \\
 &= \tan^{-1} \left(\frac{9}{13} \right) \\
 &= \sin^{-1} \left(\frac{9}{\sqrt{(13)^2 + 9^2}} \right) \\
 &= \sin^{-1} \left(\frac{9}{\sqrt{250}} \right) \\
 &= \sin^{-1} \left(\frac{9}{5\sqrt{10}} \right)
 \end{aligned}$$

Thus, the value of $\alpha - \beta$ is equal to $\sin^{-1} \left(\frac{9}{5\sqrt{10}} \right)$.

28. The mean and variance of seven observations are 8 and 16, respectively. If 5 of the observation are 2,4,10,12,14, then the product of the remaining two observations is :

- A. 49 B. 40
C. 45 D. 48

Ans. D

Explanation:

Let unknown observation are x_1 & x_2

$$\frac{2 + 4 + 10 + 12 + 14 + x_1 + x_2}{7} = 8$$

$$x_1 + x_2 = 14 \dots(i)$$

$$\frac{1}{7} \sum x_i^2 - (\bar{x})^2 = 16 \text{ (given)}$$

$$\frac{1}{7} (2^2 + 4^2 + 10^2 + 12^2 + 14^2 + x_1^2 + x_2^2) - (8)^2 = 16$$

$$x_1^2 + x_2^2 = 100$$

$$(x_1 + x_2)^2 - 2x_1x_2 = 100$$

$$2x_1x_2 = 96$$

$$x_1x_2 = 48$$

29. $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}}$ equal:

A. $\sqrt{2}$

B. $4\sqrt{2}$

C. $2\sqrt{2}$

D. 4

Ans.

B

Explanation:

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}} \\ & \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{\sqrt{2} - \sqrt{1 + \cos x}} \right) \left(\frac{\sqrt{2} + \sqrt{1 + \cos x}}{\sqrt{2} + \sqrt{1 + \cos x}} \right) \\ & \lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{1 - \cos x} \right) (\sqrt{2} + \sqrt{1 + \cos x}) \\ & \lim_{x \rightarrow 0} \frac{\sin^2 x}{2 \sin^2 \left(\frac{x}{2} \right)} (\sqrt{2} + \sqrt{1 + \cos x}) \\ & \lim_{x \rightarrow 0} \frac{1}{2} \left(\frac{\sin x}{x} \right)^2 \times x^2 \times \left(\frac{x/2}{\sin x/2} \right) \times \frac{1}{\left(\frac{x^2}{4} \right)} \times (\sqrt{2} + \sqrt{1 + \cos x}) \\ & = \frac{1}{2} \times 4 \times 2\sqrt{2} = 4\sqrt{2} \end{aligned}$$

30. If $f(x) = \log_e \left(\frac{1-x}{1+x} \right)$, $|x| < 1$, then

$f\left(\frac{2x}{1+x^2}\right)$ is equal to :

A. $(f(x))^2$

B. $2f(x)$

C. $2f(x^2)$

D. $-2f(x)$

Ans. B

Explanation:

$$f(x) = \log_e \left(\frac{1-x}{1+x} \right), |x| < 1$$

$$f\left(\frac{2x}{1+x^2}\right) = \log_e \left(\frac{1 - \frac{2x}{1+x^2}}{1 + \frac{2x}{1+x^2}} \right)$$

$$= \log_e \left(\frac{(1-x)^2}{(1+x)^2} \right)$$

$$= 2 \log_e \left(\frac{1-x}{1+x} \right) \quad \left[\because \log_e |A|^m = m \log_e |A| \right]$$

$$\left[\because f(x) = \log_e \left(\frac{1-x}{1+x} \right) \right]$$

$$= 2f(x)$$